Pandas1

November 3, 2026

data type specific to pandas

series dataframe (data.dataframe for new df)

[ ]:
```
import pandas as pd
df = pd.read_csv('president_heights1.csv')
```

[5]:
```
df.head
```

[5]:
```
<bound method NDFrame.head of     order                   name  height(cm)
0      16          Abraham Lincoln       193.0
1       7           Andrew Jackson       185.0
2      17           Andrew Johnson       178.0
3      44             Barack Obama         NaN
4      23        Benjamin Harrison       168.0
5      42             Bill Clinton       188.0
6      30          Calvin Coolidge       178.0
7      21        Chester A. Arthur       183.0
8      34     Dwight D. Eisenhower       179.0
9      32    Franklin D. Roosevelt       188.0
10     14          Franklin Pierce         NaN
11     41        George H. W. Bush       188.0
12     43           George W. Bush       182.0
13      1        George Washington       189.0
14     38              Gerald Ford         NaN
15     33          Harry S. Truman       175.0
16     31           Herbert Hoover       182.0
17     20        James A. Garfield         NaN
18     15           James Buchanan       183.0
19     11            James K. Polk       173.0
20      4            James Madison       163.0
21      5             James Monroe       183.0
22     39             Jimmy Carter       177.0
23      2               John Adams       170.0
24     35          John F. Kennedy         NaN
25      6        John Quincy Adams       171.0
26     10                John Tyler       183.0
27     36        Lyndon B. Johnson       193.0
28      8         Martin Van Buren       168.0
```

```
29     13        Millard Fillmore       175.0
30     37           Richard Nixon       182.0
31     40           Ronald Reagan       185.0
32     19     Rutherford B. Hayes       174.0
33     26      Theodore Roosevelt       178.0
34      3        Thomas Jefferson       189.0
35     18        Ulysses S. Grant       173.0
36     29       Warren G. Harding       183.0
37      9  William Henry Harrison       173.0
38     27     William Howard Taft       182.0
39     25        William McKinley       170.0
40     28          Woodrow Wilson       180.0
41     12          Zachary Taylor       173.0>
```

[ ]:
```
df.shape
```

[ ]:
```
(42, 3)
```

[ ]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 42 entries, 0 to 41
Data columns (total 3 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   order       42 non-null     int64
 1   name        42 non-null     str
 2   height(cm)  37 non-null     float64
dtypes: float64(1), int64(1), str(1)
memory usage: 1.7 KB
```

[ ]:
```
df.describe()
```

[ ]:
```
           order  height(cm)
count  42.000000   37.000000
mean   22.476190  179.378378
std    13.152461    7.361127
min     1.000000  163.000000
25%    11.250000  173.000000
50%    22.000000  180.000000
75%    33.750000  183.000000
max    44.000000  193.000000
```

[ ]:
```
df.head(5)
```

[ ]:
```
   order              name  height(cm)
0     16    Abraham Lincoln       193.0
1      7     Andrew Jackson       185.0
```

```
2     17     Andrew Johnson       178.0
3     44       Barack Obama         NaN
4     23  Benjamin Harrison       168.0
```

[ ]:
```
df.iloc[0:5]
```

[ ]:
```
   order              name  height(cm)
0     16    Abraham Lincoln       193.0
1      7     Andrew Jackson       185.0
2     17     Andrew Johnson       178.0
3     44       Barack Obama         NaN
4     23  Benjamin Harrison       168.0
```

[ ]:
```
df.loc[3]
```

[ ]:
```
order                  44
name          Barack Obama
height(cm)             NaN
Name: 3, dtype: object
```

[ ]:
```
df.iloc[[0,9,5,8]]
```

[ ]:
```
   order                  name  height(cm)
0     16        Abraham Lincoln       193.0
9     32  Franklin D. Roosevelt       188.0
5     42          Bill Clinton       188.0
8     34   Dwight D. Eisenhower       179.0
```

[ ]:
```
df.iloc[[0,4,5,5,6,8,4],[1,2]]
```

[ ]:
```
                  name  height(cm)
0       Abraham Lincoln       193.0
4     Benjamin Harrison       168.0
5          Bill Clinton       188.0
5          Bill Clinton       188.0
6       Calvin Coolidge       178.0
8  Dwight D. Eisenhower       179.0
4     Benjamin Harrison       168.0
```

[19]:
```
df.iloc[:,[1,2]]
```

[19]:
```
                     name  height(cm)
0          Abraham Lincoln       193.0
1           Andrew Jackson       185.0
2           Andrew Johnson       178.0
3             Barack Obama         NaN
4        Benjamin Harrison       168.0
5             Bill Clinton       188.0
```

```
6          Calvin Coolidge       178.0
7        Chester A. Arthur       183.0
8      Dwight D. Eisenhower      179.0
9     Franklin D. Roosevelt      188.0
10          Franklin Pierce        NaN
11        George H. W. Bush       188.0
12           George W. Bush       182.0
13        George Washington       189.0
14              Gerald Ford        NaN
15          Harry S. Truman       175.0
16           Herbert Hoover       182.0
17        James A. Garfield         NaN
18           James Buchanan       183.0
19            James K. Polk       173.0
20            James Madison       163.0
21             James Monroe       183.0
22             Jimmy Carter       177.0
23               John Adams       170.0
24          John F. Kennedy         NaN
25        John Quincy Adams       171.0
26               John Tyler       183.0
27        Lyndon B. Johnson       193.0
28         Martin Van Buren       168.0
29         Millard Fillmore       175.0
30            Richard Nixon       182.0
31            Ronald Reagan       185.0
32     Rutherford B. Hayes        174.0
33      Theodore Roosevelt        178.0
34         Thomas Jefferson       189.0
35         Ulysses S. Grant       173.0
36       Warren G. Harding        183.0
37  William Henry Harrison        173.0
38     William Howard Taft        182.0
39         William McKinley       170.0
40           Woodrow Wilson       180.0
41           Zachary Taylor       173.0
```

[ ]:
```
df.iloc[:20:2]
```

[ ]:
```
    order                 name  height(cm)
0      16       Abraham Lincoln       193.0
2      17         Andrew Johnson      178.0
4      23      Benjamin Harrison      168.0
6      30        Calvin Coolidge      178.0
8      34   Dwight D. Eisenhower      179.0
10     14        Franklin Pierce        NaN
12     43         George W. Bush      182.0
```

```
14     38           Gerald Ford         NaN
16     31        Herbert Hoover       182.0
18     15        James Buchanan       183.0
```

[ ]:
```
df[df['height(cm)'].isnull()]
```

[ ]:
```
    order               name  height(cm)
3      44       Barack Obama         NaN
10     14    Franklin Pierce         NaN
14     38        Gerald Ford         NaN
17     20  James A. Garfield         NaN
24     35    John F. Kennedy         NaN
```

[ ]:
```
df[df['height(cm)'].notnull()]
```

[ ]:
```
    order                   name  height(cm)
0      16         Abraham Lincoln       193.0
1       7          Andrew Jackson       185.0
2      17          Andrew Johnson       178.0
4      23       Benjamin Harrison       168.0
5      42            Bill Clinton       188.0
6      30         Calvin Coolidge       178.0
7      21       Chester A. Arthur       183.0
8      34    Dwight D. Eisenhower       179.0
9      32   Franklin D. Roosevelt       188.0
11     41       George H. W. Bush       188.0
12     43          George W. Bush       182.0
13      1       George Washington       189.0
15     33         Harry S. Truman       175.0
16     31          Herbert Hoover       182.0
18     15          James Buchanan       183.0
19     11           James K. Polk       173.0
20      4           James Madison       163.0
21      5            James Monroe       183.0
22     39            Jimmy Carter       177.0
23      2              John Adams       170.0
25      6       John Quincy Adams       171.0
26     10              John Tyler       183.0
27     36       Lyndon B. Johnson       193.0
28      8        Martin Van Buren       168.0
29     13        Millard Fillmore       175.0
30     37           Richard Nixon       182.0
31     40           Ronald Reagan       185.0
32     19     Rutherford B. Hayes       174.0
33     26      Theodore Roosevelt       178.0
34      3         Thomas Jefferson       189.0
35     18         Ulysses S. Grant       173.0
```

```
36     29       Warren G. Harding       183.0
37      9  William Henry Harrison       173.0
38     27     William Howard Taft       182.0
39     25        William McKinley       170.0
40     28          Woodrow Wilson       180.0
41     12          Zachary Taylor       173.0
```

[24]:
```
clean_df = df.dropna()
print(f" DF : {df.shape}")
print(f" Clean_DF : {clean_df.shape}")
```

```
 DF : (42, 3)
 Clean_DF : (37, 3)
```

[ ]:
```
Ffill = df.ffill()
print(f" Shape : {Ffill.shape}")
print(Ffill[df['height(cm)'].isnull()])
```

```
 Shape : (42, 3)
    order               name  height(cm)
3      44       Barack Obama       178.0
10     14    Franklin Pierce       188.0
14     38        Gerald Ford       189.0
17     20  James A. Garfield       182.0
24     35    John F. Kennedy       170.0
```

[ ]:
```
early_tall = df[(df['height(cm)'] > 185) & (df['order'] <= 20)]
early_tall
```

[ ]:
```
    order               name  height(cm)
0      16    Abraham Lincoln       193.0
13      1  George Washington       189.0
34      3   Thomas Jefferson       189.0
```

[ ]:
```
max_df = df[df['height(cm)'] == df['height(cm)'].max()]
max_df
```

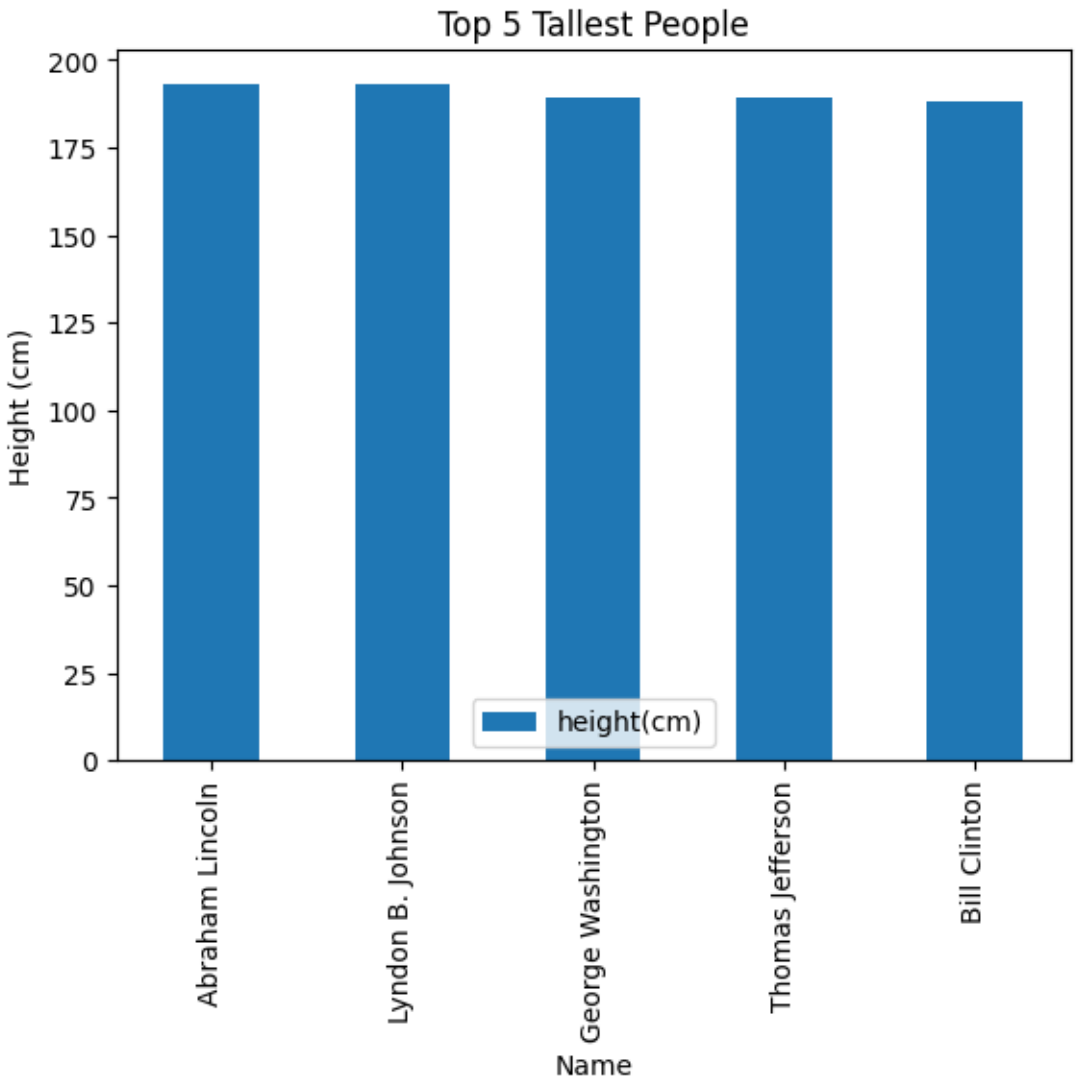
[ ]:
```
    order               name  height(cm)
0      16    Abraham Lincoln       193.0
27     36  Lyndon B. Johnson       193.0
```

[ ]:
```
tall3 = df.nlargest(3,'height(cm)')
tall3
```

[ ]:
```
    order               name  height(cm)
0      16    Abraham Lincoln       193.0
27     36  Lyndon B. Johnson       193.0
13      1  George Washington       189.0
```

[38]:
```
top5 = df.nlargest(5,'height(cm)')
top5.plot.bar(x='name', y='height(cm)', title='Top 5 Tallest People',␣
↪ylabel='Height (cm)', xlabel='Name')
```

[38]:
```
<Axes: title={'center': 'Top 5 Tallest People'}, xlabel='Name', ylabel='Height
(cm)'>
```



[ ]: